

EE 596

Project 1

Design of fully differential Opamp and switched-capacitor integrator

Pts: 100, Due: 1/20/2019 @ 11:59PM Blackboard

Part 1: Design a 2-stage fully-differential opamp with CMFB (continuous-time or switched-capacitor) using a telescopic or folded cascode input stage. The opamp should meet the following specifications and has to be simulated in Cadence. You can choose the type of lead-compensation method. This amplifier will drive a capacitive load of 1 pF (open loop).

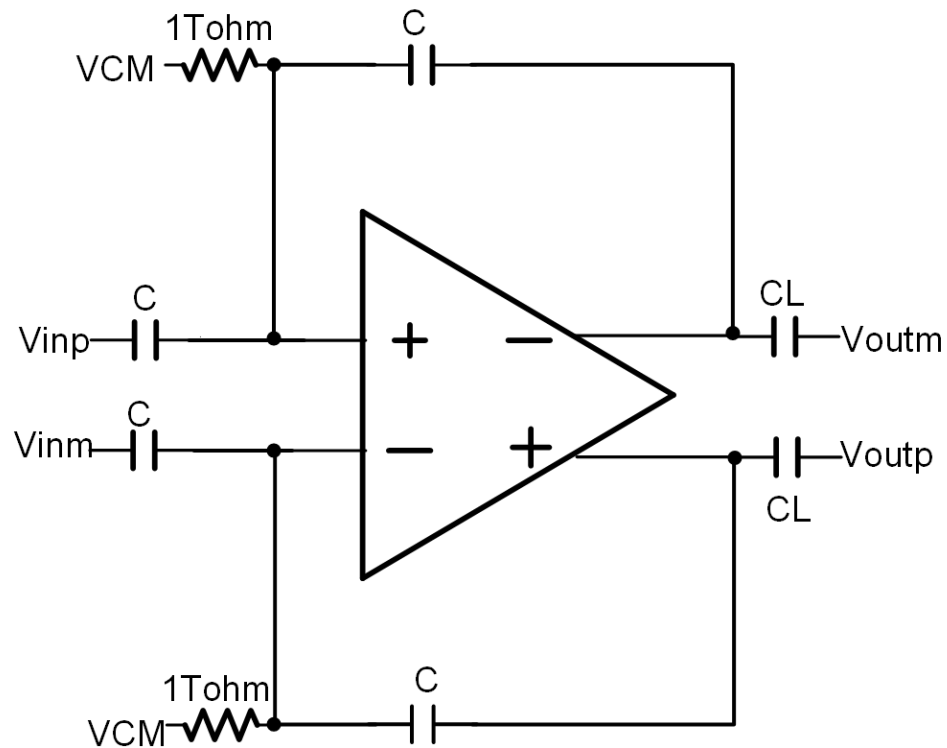
Additional material: Use the attached supplements under Content/Project/Project1 folder for simulating closed-loop gain in DM and CM.

Add notes in the adjacent column to state your assumptions as needed.

Specifications table (20 pts):

Parameter	Unit	Range	Simulation Results	Notes
Supply voltage	V	1.2		
Power dissipation	mW	<40		
DC Gain	dB	60		
fT	MHz	1200		1 pF load on each output
Phase Margin	°	>60		
Output range over which CMFB operates	V _{pk-pk,diff}	>0.4		
Input CM Range	V _{pk-pk,diff}	>0.4		
Power Supply Rejection Ratio	dB	>20 dB close-in >40 dB far-out		
Slew Rate	V/μs	>80		
Closed-loop DM gain	dB	>60		
Closed-loop DM Phase	°	>55		
Closed-loop UGB	MHz	>500		
Closed-loop CM Gain	dB			
Closed-loop CM Phase	°	>55		
Closed-loop CM UGB	MHz	>100		

Use the following circuit to measure closed-loop differential-mode and common-mode gains. The $1\text{ T}\Omega$ resistor sets the initial values at the inputs of the opamps to V_{CM} . $C=10\text{ pF}$, $CL = 1\text{ pF}$.



Supplementary Materials:

Find attached a guide to run simulations for measuring the closed-loop DM and CM gain of a fully-differential opamp in a capacitive feedback configuration. (thanks to Prof. Vishal Saxena at University of Idaho).